Square — generic $A \cdot B$ (4-bit nibbles)

the three cases by number of unsigned operands — no specific mode

Setup 4-bit nibble: signed $\in [-8, 7]$, unsigned $\in [0, 15]$. Bias an *unsigned* nibble by -8 (flip its MSB) $\Rightarrow [-8, 7]$.

$$(A{+}B)^2 = A^2 + B^2 + 2AB \qquad \Longrightarrow \qquad 2AB = (A{+}B)^2 - A^2 - B^2.$$

Case 0 — both signed (no bias)

$$\boxed{(1) \quad AB = \tfrac{1}{2}\big[(A{+}B)^2 - A^2 - B^2\big]} \qquad C = 0$$

Case 1 — one unsigned (A unsigned $\to A{-}8$; B signed)

$$\big((A{-}8){+}B\big)^2 = A^2 + B^2 + 2AB - 16A - 16B + 64$$

$$(2) \quad 2AB = \big((A{-}8){+}B\big)^2 - A^2 - B^2 + 16A + 16B - 64$$

substitute $-A^2 = -(A{-}8)^2 + 64 - 16A$, $-B^2 = -(B{-}8)^2 + 64 - 16B$ (linear terms cancel):

$$\boxed{(3) \quad AB = \tfrac{1}{2}\Big[\underbrace{\big((A{-}8){+}B\big)^2}_{\text{5-bit sq}} - \underbrace{(A{-}8)^2}_{\text{4-bit sq}} - \underbrace{(B{-}8)^2}_{\text{5-bit sq}} + 64\Big]} \qquad C = 64$$

Case 2 — both unsigned ($A \to A{-}8$, $B \to B{-}8$)

$$\big((A{-}8){+}(B{-}8)\big)^2 = A^2 + B^2 + 2AB - 32A - 32B + 256$$

$$(4) \quad 2AB = \big((A{-}8){+}(B{-}8)\big)^2 - A^2 - B^2 + 32A + 32B - 256$$

substitute $-A^2 = -(A{-}16)^2 + 256 - 32A$, $-B^2 = -(B{-}16)^2 + 256 - 32B$ (linear terms cancel):

$$\boxed{(5) \quad AB = \tfrac{1}{2}\Big[\underbrace{\big((A{-}8){+}(B{-}8)\big)^2}_{\text{5-bit sq}} - \underbrace{(A{-}16)^2}_{\text{5-bit sq}} - \underbrace{(B{-}16)^2}_{\text{5-bit sq}} + 256\Big]} \qquad C = 256$$

Complete table (all four combos) au, bu = A/B unsigned flags, $S = 8(au{+}bu)$, $C = S^2$. Result $= \frac{1}{2}(\text{PE} - \alpha - \beta + C)$.

au	bu	(A,B)	S	f	PE	$\alpha = (A{-}S)^2$	$\beta = (B{-}S)^2$
0	0	(s,s)	0	(1)	$(A{+}B)^2$ 5b	A^2 4b	B^2 4b
1	0	(u,s)	8	(3)	$\big((A{-}8){+}B\big)^2$ 5b	$(A{-}8)^2$ 4b	$(B{-}8)^2$ 5b
0	1	(s,u)	8	(3)	$\big(A{+}(B{-}8)\big)^2$ 5b	$(A{-}8)^2$ 5b	$(B{-}8)^2$ 4b
1	1	(u,u)	16	(5)	$\big((A{-}8){+}(B{-}8)\big)^2$ 5b	$(A{-}16)^2$ 5b	$(B{-}16)^2$ 5b

$C = 0, 64, 64, 256$ respectively. PE-arg $\in [-16, 14]$, α, β-arg $\in [-16, 7]$ — all fit one **5-bit-in / 9-bit-out** square (rows 10/01: α, β are both $A{-}8/B{-}8$ but the widths swap).

Hardware Dispatcher centers each nibble (A_c = flip A_3 if au; B_c = flip B_3 if bu) — shared by PE and α/β. Each generator adds the *removed* operand's -8 on its own input.

au	bu	A_c (disp)	α-arg	B_c (disp)	β-arg
0	0	A	$\{A_3\,A_3\,A_{2:0}\}{=}A$	B	$\{B_3\,B_3\,B_{2:0}\}{=}B$
1	0	$A{-}8$	$\{\overline{A_3}\,\overline{A_3}\,A_{2:0}\}{=}A{-}8$	B	$\{1\,\overline{B_3}\,B_{2:0}\}{=}B{-}8$
0	1	A	$\{1\,\overline{A_3}\,A_{2:0}\}{=}A{-}8$	$B{-}8$	$\{\overline{B_3}\,\overline{B_3}\,B_{2:0}\}{=}B{-}8$
1	1	$A{-}8$	$\{1\,A_3\,A_{2:0}\}{=}A{-}16$	$B{-}8$	$\{1\,B_3\,B_{2:0}\}{=}B{-}16$

Generator remap (bits [2:0] pass through): $\alpha_4{=}bu \vee (A_3{\oplus}au)$, $\alpha_3{=}A_3{\oplus}au{\oplus}bu$; $\beta_4{=}au \vee (B_3{\oplus}bu)$, $\beta_3{=}B_3{\oplus}au{\oplus}bu$.

Checks (3) $A{=}15, B{=}7$: $\frac{1}{2}[14^2 - 7^2 - (-1)^2 + 64] = \frac{1}{2} \cdot 210 = 105 = 15 \cdot 7$; (5) $A{=}15, B{=}15$: $\frac{1}{2}[14^2 - (-1)^2 - (-1)^2 + 256] = \frac{1}{2} \cdot 450 = 225 = 15 \cdot 15$.